

Design of a 4-bit NOT Gate Using VHDL and Logisim Simulation

1. Objective

Describe a 4-bit NOT gate using VHDL, construct the circuit in Logisim-evolution, and verify that the output is inverted bit-by-bit relative to the input.

2. VHDL Code

```
1  LIBRARY ieee;
2  USE ieee.std_logic_1164.all;
3
4  ENTITY NOT_GATE IS
5  PORT (
6    a : IN  std_logic_vector(3 DOWNTO 0);
7    y : OUT std_logic_vector(3 DOWNTO 0)
8  );
9  END NOT_GATE;
10
11 -----
12 --Complete your VHDL description below
13 -----
14
15 ARCHITECTURE behavior OF NOT_GATE IS
16 BEGIN
17     y <= NOT a;
18
19 END behavior;
```

Fig 1. VHDL Code of NOT_GATE

In the ENTITY, a 4-bit input port `a` and a 4-bit output port `y` are declared as `std\_logic\_vector (3 DOWNTO 0)`.

In the ARCHITECTURE, a single concurrent signal assignment statement, `y <= NOT a;`, inverts all bits of the input and passes them to the output.

3. Circuit & Simulation Result

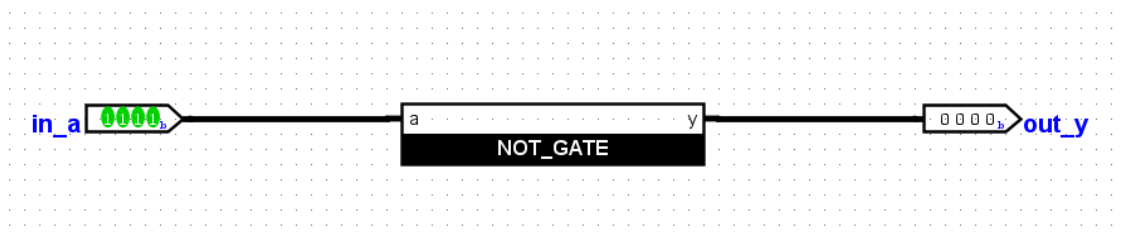


Fig 2. Input 1111 → Output 0000



Fig 3. Input 0000 → Output 1111

Cases	Input 'a'	Output 'b'
1	1111	0000
2	0000	1111

Table 1. Result Overview

In both cases, the output matched the value obtained by inverting each bit of the input.